

Optimizing Renewable Energy Production With Constraints and Subsidies

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Introduction

The objective of this report is to solve a revenue maximization problem for a firm that produces and sells renewable energy given its inputs and capacity constraints, receiving subsidies from the government for every unit of energy sale. This is interesting because in a lot of developing nations, the governments initially provide subsidies for the green energy-producing firms to build a reputation for the country, which helps them attract foreign investments, earn favorable treatment in trade agreements and create a more competitive and safe energy mix for their nations. However, this phenomenon is not exclusive to developing nations; the US Energy Information Administration (2023) reported that annual renewable energy producers receive a sizable subsidy in the United States and that these subsidies related to natural gas and petroleum shifted from a revenue-generating trend to becoming a net cost for the federal government in recent years. On a similar note, the Environmental and Energy Study Institute (2019, 2021) has recently emphasized the need to reconsider and potentially phase out fossil fuel subsidies, given the mature and profitable state of the domestic fossil fuel industries and the growing competitiveness of clean and renewable alternatives, with the End Polluter Welfare Act and the Clean Energy for America Act being two examples of bills that the 117th Congress presented as potential steps towards reforming fossil fuel subsidies towards more environmentally friendly sources.

In fact, according to a report by the Grantham Research Institute on Climate Change and the Environment (2018), it is noted that renewable energy producers receive more financial support than fossil and nuclear power producers, which is justified by the fact that renewable energy sources are becoming more cost-competitive with fossil fuels and nuclear power and that market failures such as the discrepancy in prices with fossil fuels occur when environmental costs are not taken into account. The article also mentioned how important a role these subsidies played in spurring innovation and increasing the use of renewable energy during the transitional phase. This, in turn, led to a notable rise in the deployment of renewable energy and a sharp decline in the cost of technologies such as offshore wind.

Some have, meanwhile, also criticized energy subsidies generally. The International Monetary Fund (2013) asserts that all energy subsidies impede private investment, distort resource allocation, contribute to fiscal imbalances, divert funds from important public spending priorities, and primarily benefit higher-income households, which both contribute to and reinforce economic inequality. In line with the worry, Sovacool (2017) noted that these subsidies come at a net cost to society in the form of increasing waste, government deficits, energy shortages, and crime associated with the illegal fuel trade, among other things.

This study sought to determine the complex relationships between market prices, subsidies, and input costs and the quantity of labor and technology used, as well as the factors that influence these relationships. While the optimal input demands clearly had a negative relationship with their respective prices, the other correlations weren't as straightforward.

Model

In this paper, a simple model is assumed for a monopolistic firm that has been granted the sole right and license by the government to produce renewable energy (q) and sell per kilowatt-hour (kWh) of it at price p , which is also determined by the government, while the government provides the firm subsidy (s) per unit of sale as an incentive. The firm's objective is to maximize revenue by optimizing the level of its inputs, Labor (L) and Technology (T), where wage (w) and rent (r) are the costs associated with them, respectively, given a total budget of B to spend on the two input factors. L here refers to a skilled workforce, and T is the availability of technology, which includes solar-powered photovoltaic cells and panels, gearboxes, turbines, and generators for wind, hydro, and tidal energy, as well as controlled incinerators for biomass. The firm tries to find a bundle of inputs it can afford to maximize its profit.

It's also assumed that the marginal product of technology increases as more labor is employed, as more skilled workers can make better use of the existing technological resources available and generate better innovations and developments in this sector. In a similar vein, the marginal product of labor also increases as more technology is available to work with.

The objective function is: $\max R(q) = (p + s)q$ subject to $rT + wL = B$ and $q = q(T, L)$

and so, the Lagrangian is: $\mathcal{L}(B, p, s, r, w) = (p + s)q(T, L) + \lambda (B + s - rT - wL)$.

Here, the Lagrange multiplier (λ) is the marginal benefit (additional revenue received/lost for changing B by one dollar) of the budget which is the rate the revenue would change at if the budget was relaxed by a small margin.

First Order Conditions (FOCs)

In order to find the optimal levels of λ , T , and L , the first order conditions of the Lagrangian with respect to λ , T , and L are set to zero. This is because FOC here represents the rate of change of the Lagrangian, and the rate of change becomes zero when the slope becomes horizontal which only happens at the optimum (here, maximum) point.

$$(i) \mathcal{L} = B + s - rT - wL = 0 \Rightarrow B + s = rT - wL$$

This implies that the firm is exhausting its entire budget investing in T and L . This is also the equation of the firm's budget line.

$$(ii) \mathcal{L}_T = (p + s)q_T - \lambda r = 0 \Rightarrow (p + s)q_T = \lambda r$$

For the optimal T , it's required that the firm keeps renting T until the marginal value product of doing so just equal to the marginal cost (r). λr here is the marginal revenue forgone by spending r on another unit of T .

$$(iii) \mathcal{L}_L = (p + s)q_L - \lambda w = 0 \Rightarrow (p + s)q_L = \lambda w$$

For the optimal L , it's required that the firm keeps hiring L until the marginal value product of doing so is just equal to the marginal cost (w). λw here is the marginal revenue forgone by spending w on another unit of L .

In the equations above, if the value of an exogenous variable changes on either side, the other relevant parameters must change accordingly until the equations are satisfied again.

Graphical Representation

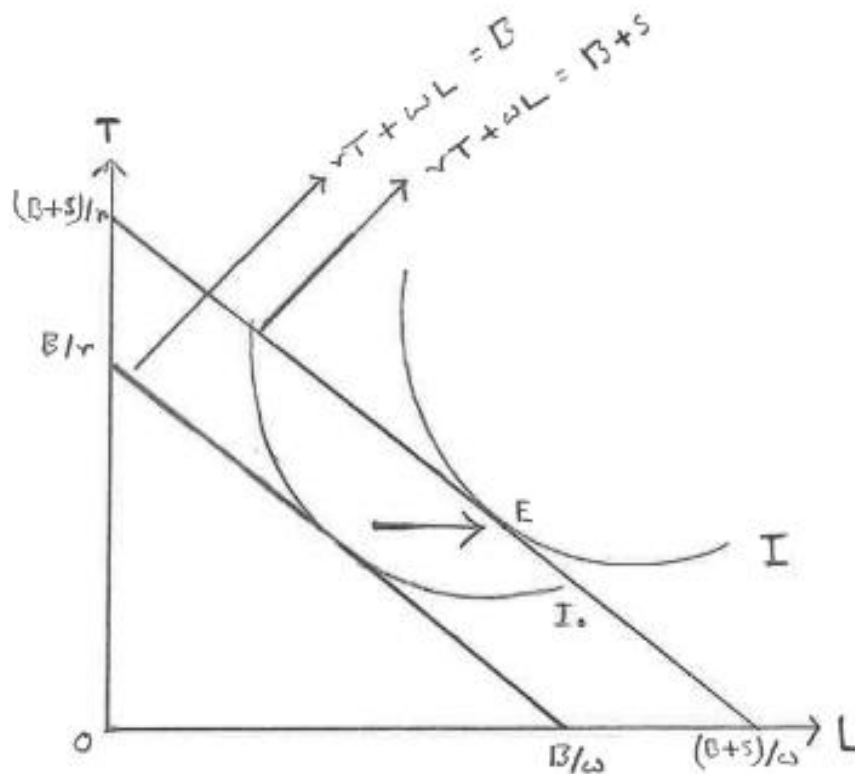
Rearranging (i) we find,

$$(a) \quad T = \frac{B+s}{r} + \left(\frac{w}{r}\right)L$$

$$(b) \quad L = \frac{B+s}{w} + \left(\frac{r}{w}\right)T$$

Setting $L = 0$ in (a) $\Rightarrow T^{\text{intercept}} = \frac{B+s}{r}$ and similarly $T = 0$ in (b) $\Rightarrow L^{\text{intercept}} = \frac{B+s}{w}$.

Here, the marginal rate of technical substitution, $-\frac{(p+s)q_T}{(p+s)q_L}$ equals $-\frac{r}{w}$, the input price ratio, at the tangency point (E) of Isoquant (I) and the budget line, $B + s = rT - wL$. However, in the absence of the subsidy, the budget line would be $B = rT + wL$ and that would have a tangency with I_0 at the left the new budget line. The subsidy, therefore, allows the firm to employ more of both its inputs to maximize revenue.



Since the choice variables (λ^*, T^*, L^*) are all functions of $B, p, s, r,$ and w , any change in the exogenous variables can result into different optimal bundles (λ^*, T^*, L^*) . We know from IFT that FOCs implicitly define the endogenous variables as functions of the exogenous variables, $\lambda^*(B, p, s, r, w), T^*(B, p, s, r, w), L^*(B, p, s, r, w)$ and from IFR that the comparative statics turn out to be a linear system of the form $Ax=b$ where A is the bordered Hessian H^B , x is the comp stats and b is the negative of partial derivatives of the FOCs with respect to the pertinent exogenous parameters.

Second Order Condition (SOC)

$$|H^B| = \begin{vmatrix} L_{\lambda\lambda} & L_{\lambda T} & L_{\lambda L} \\ L_{T\lambda} & L_{TT} & L_{TL} \\ L_{L\lambda} & L_{LT} & L_{LL} \end{vmatrix} = \begin{vmatrix} 0 & -r & -w \\ -r & (p+s)q_{TT} & (p+s)q_{TL} \\ -w & (p+s)q_{LT} & (p+s)q_{LL} \end{vmatrix}$$

$$= r(p+s)(wq_{TL} - rq_{LL}) - w(p+s)(wq_{TT} - rq_{LT})$$

$$= -r^2(p+s)q_{LL} - w^2(p+s)q_{TT} + 2rw(p+s)q_{TL} \text{ (young's theorem)}$$

For an optimum, SOC needs to be positive.

$$-r^2(p+s)q_{LL} - w^2(p+s)q_{TT} + 2rw(p+s)q_{TL} > 0 \Rightarrow -r^2(p+s)q_{LL} - w^2(p+s)q_{TT} > -2rw(p+s)q_{TL}$$

By the law of diminishing marginal returns, we know that $q_{LL} < 0$ and $q_{TT} < 0$ and given that p , s , r , and w are all positive, the left side of the inequality is positive. So, the sufficient condition (SOC > 0) is satisfied when q_{TL} is zero or positive and if it's negative, the magnitude of the left side must be greater than the magnitude of the right side. However, the model already assumes $q_{TL} > 0$ (from the introduction) and so SOC is satisfied.

$$\text{For a change in B: } \begin{bmatrix} 0 & -r & -w \\ -r & (p+s)q_{TT} & (p+s)q_{TL} \\ -w & (p+s)q_{LT} & (p+s)q_{LL} \end{bmatrix} \begin{bmatrix} \lambda_B \\ T_B \\ L_B \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{For a change in p: } \begin{bmatrix} 0 & -r & -w \\ -r & (p+s)q_{TT} & (p+s)q_{TL} \\ -w & (p+s)q_{LT} & (p+s)q_{LL} \end{bmatrix} \begin{bmatrix} \lambda_p \\ T_p \\ L_p \end{bmatrix} = \begin{bmatrix} 0 \\ -q_T \\ -q_L \end{bmatrix}$$

$$\text{For a change in s: } \begin{bmatrix} 0 & -r & -w \\ -r & (p+s)q_{TT} & (p+s)q_{TL} \\ -w & (p+s)q_{LT} & (p+s)q_{LL} \end{bmatrix} \begin{bmatrix} \lambda_s \\ T_s \\ L_s \end{bmatrix} = \begin{bmatrix} -1 \\ -q_T \\ -q_L \end{bmatrix}$$

$$\text{For a change in r: } \begin{bmatrix} 0 & -r & -w \\ -r & (p+s)q_{TT} & (p+s)q_{TL} \\ -w & (p+s)q_{LT} & (p+s)q_{LL} \end{bmatrix} \begin{bmatrix} \lambda_r \\ T_r \\ L_r \end{bmatrix} = \begin{bmatrix} T \\ \lambda \\ 0 \end{bmatrix}$$

$$\text{For a change in w: } \begin{bmatrix} 0 & -r & -w \\ -r & (p+s)q_{TT} & (p+s)q_{TL} \\ -w & (p+s)q_{LT} & (p+s)q_{LL} \end{bmatrix} \begin{bmatrix} \lambda_w \\ T_w \\ L_w \end{bmatrix} = \begin{bmatrix} L \\ 0 \\ \lambda \end{bmatrix}$$

Crammer's Rule

$$|H\lambda_B| = \begin{vmatrix} -1 & -r & -w \\ 0 & (p+s)q_{TT} & (p+s)q_{TL} \\ 0 & (p+s)q_{LT} & (p+s)q_{LL} \end{vmatrix} = (p+s)^2 (q_{TL}^2 - q_{TT} q_{LL})$$

$$|HT_B| = \begin{vmatrix} 0 & -1 & -w \\ -r & 0 & (p+s)q_{TL} \\ -w & 0 & (p+s)q_{LL} \end{vmatrix} = (p+s) (w q_{TL} - r q_{LL})$$

$$|HL_B| = \begin{vmatrix} 0 & -r & -1 \\ -r & (p+s)q_{TT} & 0 \\ -w & (p+s)q_{LT} & 0 \end{vmatrix} = (p+s) (r q_{TL} - w q_{TT})$$

$$\begin{aligned} |H\lambda_p| &= \begin{vmatrix} 0 & -r & -w \\ -q_T & (p+s)q_{TT} & (p+s)q_{TL} \\ -q_L & (p+s)q_{LT} & (p+s)q_{LL} \end{vmatrix} \\ &= (p+s) [(p+s) q_T (q_{TT} q_{LL} - q_{TL}^2) - q_L (w q_{TT} - r q_{TL})] \end{aligned}$$

$$|HT_p| = \begin{vmatrix} 0 & 0 & -w \\ -r & -q_T & (p+s)q_{TL} \\ -w & -q_L & (p+s)q_{LL} \end{vmatrix} = w (w q_T - r q_L)$$

$$|HL_p| = \begin{vmatrix} 0 & -r & 0 \\ -r & (p+s)q_{TT} & -q_T \\ -w & (p+s)q_{LT} & -q_L \end{vmatrix} = r (r q_L - w q_T)$$

$$\begin{aligned} |H\lambda_s| &= \begin{vmatrix} -1 & -r & -w \\ -q_T & (p+s)q_{TT} & (p+s)q_{TL} \\ -q_L & (p+s)q_{LT} & (p+s)q_{LL} \end{vmatrix} \\ &= (p+s) [(p+s) q_{TL}^2 - (p+s) q_{TT} q_{LL} + q_T (w q_{TL} - r q_{LL}) + q_L (w q_{TT} - r q_{TL})] \end{aligned}$$

$$|HT_s| = \begin{vmatrix} 0 & -1 & -w \\ -r & -q_T & (p+s)q_{TL} \\ -w & -q_L & (p+s)q_{LL} \end{vmatrix} = (p+s) (w q_{TL} - r q_{LL}) + w (w q_T - r q_L)$$

$$|HL_s| = \begin{vmatrix} 0 & -r & -1 \\ -r & (p+s)q_{TT} & -q_T \\ -w & (p+s)q_{LT} & -q_L \end{vmatrix} = r (r q_L - w q_T) - (p+s) (w q_{TT} - r q_{TL})$$

$$|H\lambda_r| = \begin{vmatrix} T & -r & -w \\ \lambda & (p+s)q_{TT} & (p+s)q_{TL} \\ 0 & (p+s)q_{LT} & (p+s)q_{LL} \end{vmatrix}$$

$$= T (p + s)^2 (q_{TT} q_{LL} - q_{TL}^2) - \lambda (p + s) (w q_{TL} - r q_{LL})$$

$$|HT_r| = \begin{vmatrix} 0 & T & -w \\ -r & \lambda & (p + s)q_{TL} \\ -w & 0 & (p + s)q_{LL} \end{vmatrix} = rT (p + s) q_{LL} - w [T (p + s) q_{TL} + w\lambda]$$

$$|HL_r| = \begin{vmatrix} 0 & -r & T \\ -r & (p + s)q_{TT} & \lambda \\ -w & (p + s)q_{LT} & 0 \end{vmatrix} = rw\lambda + T (p + s) (w q_{TT} - r q_{TL})$$

$$|H\lambda_w| = \begin{vmatrix} L & -r & -w \\ 0 & (p + s)q_{TT} & (p + s)q_{TL} \\ \lambda & (p + s)q_{LT} & (p + s)q_{LL} \end{vmatrix}$$

$$= L (p + s)^2 (q_{TT} q_{LL} - q_{TL}^2) + \lambda [(p + s) (w q_{TT} - r q_{TL})]$$

$$|HT_w| = \begin{vmatrix} 0 & L & -w \\ -r & 0 & (p + s)q_{TL} \\ -w & \lambda & (p + s)q_{LL} \end{vmatrix} = r [(p + s) L q_{LL} + w \lambda] - wL (p + s) q_{TL}$$

$$|HL_w| = \begin{vmatrix} 0 & -r & L \\ -r & (p + s)q_{TT} & 0 \\ -w & (p + s)q_{LT} & \lambda \end{vmatrix} = wL (p + s) q_{TT} - r [r \lambda + L (p + s) q_{TL}]$$

Solving for the Comp Stats

$$\lambda_B = \frac{(p + s)^2 (q_{TL}^2 - q_{TT} q_{LL})}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$T_B = \frac{(p + s) (w q_{TL} - r q_{LL})}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$L_B = \frac{(p + s) (r q_{TL} - w q_{TT})}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$\lambda_p = \frac{(p + s) [(p + s) q_T (q_{TT} q_{LL} - q_{TL}^2) - q_L (w q_{TT} - r q_{TL})]}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$T_p = \frac{w (w q_T - r q_L)}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$L_p = \frac{r (r q_L - w q_T)}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$\lambda_s = \frac{(p + s) [(p + s) q_{TL}^2 - (p + s) q_{TT} q_{LL} + q_T (w q_{TL} - r q_{LL}) + q_L (w q_{TT} - r q_{TL})]}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$T_s = \frac{(p + s) (w q_{TL} - r q_{LL}) + w (w q_T - r q_L)}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$L_s = \frac{r (r q_L - w q_T) - (p + s) (w q_{TT} - r q_{TL})}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$\lambda_r = \frac{T (p + s)^2 (q_{TT} q_{LL} - q_{TL}^2) - \lambda (p + s) (w q_{TL} - r q_{LL})}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$T_r = \frac{r T (p + s) q_{LL} - w [T (p + s) q_{TL} + w \lambda]}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$L_r = \frac{rw \lambda + T (p + s) (w q_{TT} - r q_{TL})}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$\lambda_w = \frac{L (p + s)^2 (q_{TT} q_{LL} - q_{TL}^2) + \lambda [(p + s) (w q_{TT} - r q_{TL})]}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$T_w = \frac{r [(p + s) L q_{LL} + w \lambda] - w L (p + s) q_{TL}}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

$$L_w = \frac{w L (p + s) q_{TT} - r [r \lambda + L (p + s) q_{TL}]}{-r^2 (p + s) q_{LL} - w^2 (p + s) q_{TT} + 2rw(p + s) q_{TL}}$$

We know that q_{TT} and q_{LL} are negative (diminishing returns) while r , w , p , s are all positive (economic intuition) and for an interior solution, λ , T , and L must also be positive. The denominator, H^B is also positive from SOC. Hence, the signs of comparative statics depend on the sign of the numerator. Since, it's already assumed that $q_{TL} > 0$, technology and labor are complements and so, analyzing the signs in the numerator, we find the following comparative statics.

Comparative Statics

$\lambda_B = (+) [0 - (-) (-)] = (-)$: As the budget increases, marginal revenue received from the additional budget decreases and vice versa.

$T_B = (+) [0 - (+) (-)] = (+)$: As the budget increases, more labor should be hired to maximize revenue and vice versa.

$L_B = (+) [0 - (+) (-)] = (+)$: As the budget increases, more technology should be rented to maximize revenue and vice versa.

$\lambda_p = (+) [(+) (+) \{(-) - 0\} - \{(+) (-) - 0\}] = (+)$: As price goes up, marginal revenue of budget goes up as well and vice versa.

$T_p = (+) [(+)(+) - (+)(+)] = (?)$: If $wq_T > rq_L$, then an increase in price increases optimum level of technology, if $wq_T < rq_L$, then the opposite relationship is achieved while if they are equal, then price change doesn't influence the optimal level of technology.

$L_p = (+) [(+)(+) - (+)(+)] = (?)$: If $wq_T > rq_L$, then an increase in price decreases optimum level of labor, if $wq_T < rq_L$, then the opposite relationship is achieved while if they are equal, then price change doesn't influence the optimal level of labor.

$\lambda_s = (+) [\{0 - (+)(-)(-)\} + (+)\{0 - (+)(-)\} + (+)\{(+)(-) - 0\}] = (?)$: If $q_T (w q_{TL} - r q_{LL}) > [q_L (w q_{TT} - r q_{TL}) - (p + s) q_{TT} q_{LL}]$, then an increase in subsidy increases the marginal revenue received from the budget, if the opposite is true, then the marginal revenue of budget falls and if the terms are equal, there is no effect.

$T_s = (+) [\{0 - (+)(-)\} + (+) \{(+)(+) - (+)(+)\}] = (?)$: If $wq_T \geq rq_L$, then a rise in subsidy influences the optimal level of technology positively, otherwise, the relationship depends on the magnitudes.

$L_s = (+) [\{(+)(+) - (+)(+)\} - (+) \{(+)(-) - 0\}] = (?)$: If $wq_T \leq rq_L$, then a rise in subsidy influences the optimal level of labor positively, otherwise, the relationship depends on the magnitudes.

$\lambda_r = [(+) (+) \{(-) (-) - 0\}] - [(+) (+) \{0 - (+) (-)\}] = (?)$: If the 1st term > the 2nd term, rent influences marginal revenue of budget positively, if equal, there is no impact, if otherwise, rent affects λ negatively.

$T_r = (+) (+) (-) - (+) [0 + (+) (+)] = (-)$: If rent increases, the firm should borrow less technology and vice versa.

$L_r = (+) + (+) (+) [(+) (-) - 0] = (?)$: If $rw\lambda > T(p + s)(w q_{TT} - r q_{TL})$, then increasing rent alters optimal labor positively, if they are equal, then there is no effect, if otherwise, the relationship is exactly the opposite.

$\lambda_w = (+) (+) [(-) (-) - 0] + (+) (+) [(+) (-) - 0] = (?)$: If the 1st term $>$ the 2nd term, wage influences marginal revenue of budget positively, if equal, there is no impact, if otherwise, it affects λ negatively.

$T_w = (+) [(+) (+) (-) + (+) (+)] - (+) (+) (+) (0) = (?)$: If $w\lambda > (P + s)L q_{LL}$, then wage influences optimal level of technology positive, if equal, there is no impact, if otherwise, it affects T negatively.

$L_w = (+) (+) (-) - (+) [(+) + (+) (+) (0)] = (-)$: If wage increases, the firm should hire less labor and vice versa.

Conclusion

The impact of prices, subsidies, and input prices on renewable energy production and revenues received from it is a compelling and complex subject. Numerous stakeholders are paying close attention to and evaluating how these green energy producers behave, as well as what variables impact their decisions, in light of the renewable energy sector's rapid rise.

The study's only definitive conclusions relate to the relationship between the budget, input prices, and demand—a larger budget allows the company to hire more inputs, while higher input prices cause the company to hire fewer inputs. The marginal return of the budget goes down as the supply of budget goes up, which also follows economic intuition. Similarly, higher prices increase the marginal return on a budget because they increase the possibility of a larger revenue margin since the firm can raise more money to boost production.

It is crucial to recognize, nonetheless, that this study has several limitations with regard to its approach and scope. One obvious drawback is that most comparative statistics depend on the magnitude of the components; in the absence of actual data, they don't lead to any meaningful conclusions. Another problem is that the firm's unit subsidy is assigned to its budget for the following period instead of the current one. As a result, the company is able to increase its input investments in the future rather than getting a budget increase and having to change production simultaneously in the current period. It is therefore recommended to conduct an extensive study in order to determine the changing dynamics of this relationship during various time periods. Therefore, the improved model should account for the dynamic changes instead of a static one, which is outside the purview of this research endeavor. Consequently, it is clear that there is a great deal of room for further research using this particular analytical methodology.

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